

Patrick PRELL

Robotics & Embedded Systems Engineer

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I grew up on a sheep farm in the Southern Tablelands. For eight years I've built hardware and software that ships to real customers. I started on ArduPilot and PX4 survey UAVs at CORDEL. That work covered bring-up, control tuning, mission planning and field flight operations. I also built an EKF that fused IMU, RTK GNSS and LiDAR to localise a survey platform outdoors.

At ResusRight I led firmware for a clinical monitor under IEC 62304 and ISO 13485, where reliability was not negotiable. I now build RF direction-finding systems and embedded Linux at DroneShield, on products that deploy to the field. I own systems end to end and mentor the engineers around me.

SKILLS

Aerial Robotics	ArduPilot and PX4 flight stacks, UAV bring-up, tuning and mission planning, GNSS/RTK navigation, field flight operations
Control & Estimation	Extended Kalman Filter, Bayesian sensor fusion, full-state feedback, observer design, fixed-rate discrete-time control loops, MATLAB
Robotics & Autonomy	Localisation and pose estimation (IMU / RTK GNSS / LiDAR), vision-based navigation, point cloud processing, ROS 2 (working knowledge)
Perception & Vision	OpenCV (C++), optical flow, 3D reconstruction from 2D imagery, calibration, ground-truth evaluation and visualisation
Embedded & Systems	Embedded Linux, bare-metal C/C++, bootloaders, STM32 / NXP / Silicon Labs / ESP32, Bluetooth LE, gRPC, Qt, Flutter, JavaScript, POSIX shell
Field & Hardware	PCB design (KiCad, Eagle), 3D CAD (PTC Creo, Fusion 360), SMD soldering and rework, 3D printing (FDM/SLA), CNC – comfortable taking hardware from prototype to production
Leadership	Led small engineering teams to on-time delivery, mentoring and university tutoring, coordinating work across production, hardware and test engineering

PROFESSIONAL EXPERIENCE

Current Nov 2023	Embedded Software Engineer DroneShield - Australia - Delivered the conducted calibration behind the angle-of-arrival estimation for a four-antenna direction-finding system (400 MHz–7 GHz). Designed simultaneous multi-channel injection through a characterised splitter, producing a per-channel correction table that removes amplitude and phase imbalance and makes bearing accuracy repeatable unit to unit. - Cut analog front-end calibration from over 12 hours to 40 minutes – a 20x improvement that removed the single largest bottleneck in production. Required a full rearchitecture and coordination across production, hardware and test engineering. - Lead the effort for cleaning up and modernising our C++ Qt integrated display codebase. Enforcing clang formatting and linting style guidelines. greatly improving code readability and reducing the bug count within this project. - Built Python (Dash/Plotly) analysis tooling for frequency sweeps and calibration results. Stitched S2P measurements across 11 filter banks into a single validated front-end response. Embedded C++ Python Rust RF Calibration Direction Finding FPGA/RFSoc Qt gRPC Embedded Linux numpy/scipy Dash/Plotly POSIX Shell Jira GitLab
May 2025 Sep 2019	Co-founder & CTO Borne Clothing - Australia Co-founded and grew a social-enterprise clothing business to a successful exit. Donated \$30,000+ to the Against Malaria Foundation – 5,000 families protected from mosquito-borne disease for three years. Business Operations Web Development Lean Growth

Sep 2023 May 2022	Embedded Systems Engineer ResusRight - Australia > Led firmware development of a manufacturing test jig, managing a small team and setting timelines that delivered on schedule and unblocked board production. > Architected a Qt application on an embedded Linux SOM, including system architecture and the BLE stack. Developed firmware for “Nemo”, a clinical resuscitation monitor now used on real patients (NXP and Silicon Labs MCUs, BLE and serial stacks). > Took over “Juno” firmware (ESP32, BLE) and designed its Flutter companion app with Firebase auth/database and iOS/Android Bluetooth stacks. > Worked to IEC 62304, IEC 60601 and ISO 13485 – safety-critical, traceable firmware in a startup moving at pace. Embedded C/C++ Embedded Linux Qt NXP μC Silicon Labs μC Espressif μC Bluetooth LE IEC62304 IEC60601 ISO13485 Flutter Firebase Python GitHub
May 2022 Apr 2018	Mechatronics Engineer CORDEL (formerly Airsight) - Australia > Worked hands-on with ArduPilot and PX4 flight stacks across the company’s survey UAV fleet – airframe bring-up, flight controller configuration and tuning, mission planning and field flight operations. > Contributed to GNSS navigation research, characterising RTK positioning accuracy and the behaviour of GNSS-based navigation under real field conditions. > Implemented an Extended Kalman Filter fusing IMU, RTK GNSS and LiDAR for pose estimation and localisation of a survey platform in unstructured outdoor environments. Required Bayesian filtering, linear algebra and careful handling of dropout and noise in real field data. > Developed pose and velocity estimation from optical flow in a C++/OpenCV pipeline. Spent a year in New York working largely unsupervised on 3D point extraction from 2D camera footage on a moving train. > Implemented a pseudo-RTOS to arbitrate commands over the serial port and designed the status indicator board – PCB, C and AVR assembly firmware. ArduPilot PX4 UAV Flight Operations GNSS/RTK Mission Planning C++ Python OpenCV Extended Kalman Filter Sensor Fusion LiDAR RTK GNSS Point Clouds Embedded C AVR Assembly RTOS KiCad Rust Jira Bitbucket

PROJECTS

Farm inspection drone in ROS 2 / PX4 SITL github.com/Prellly95/paddock Autonomous ROS 2 inspection stack for a VTOL platform, built to help my family monitor our farm. Autonomous take-off/land with fixed-wing cruise between paddocks. Missions gated on live flight conditions inferred from local weather reports. Telemetry feeds a homestead C2 dashboard for water level and asset monitoring. ROS 2 rlcpp PX4 uXRCE-DDS Gazebo EKF Guidance & Control C++ Python	2026
UVie – magnetic UV exposure tracker github.com/Prellly95/uvie-hardware , github.com/Prellly95/uvie-firmware , github.com/Prellly95/uvie-app Battery-free wearable: nRF54L15, LTR390-UV-01 sensor, solar harvesting (BQ25570 + supercapacitor). Embedded Rust firmware with BLE/NFC. SolidJS + Capacitor companion app sharing a no_std Rust wire protocol via WebAssembly. Hardware to app. Embedded Rust nRF54L15 KiCad BLE NFC Energy Harvesting SolidJS Capacitor WebAssembly	2026 – CURRENT
Two-wheeled balancing robot github.com/Prellly95/MCHA3K End-to-end build – PCB, plant model, Kalman observer and full-state feedback controller in MATLAB, implemented at 50 Hz in embedded C on an 8-bit ATmega32 (no FPU). Full-State Feedback Kalman Observer Real-Time Control Embedded C PCB Design MATLAB Eagle	2017

EDUCATION

Jul 2019 Feb 2015	B.Eng. Mechatronics (Honours) University of Newcastle
Feb 2022 Jan 2022	Vision-Based Navigation Industry Short Course - Dr Chris Renton

REFERENCES

Available on request.